

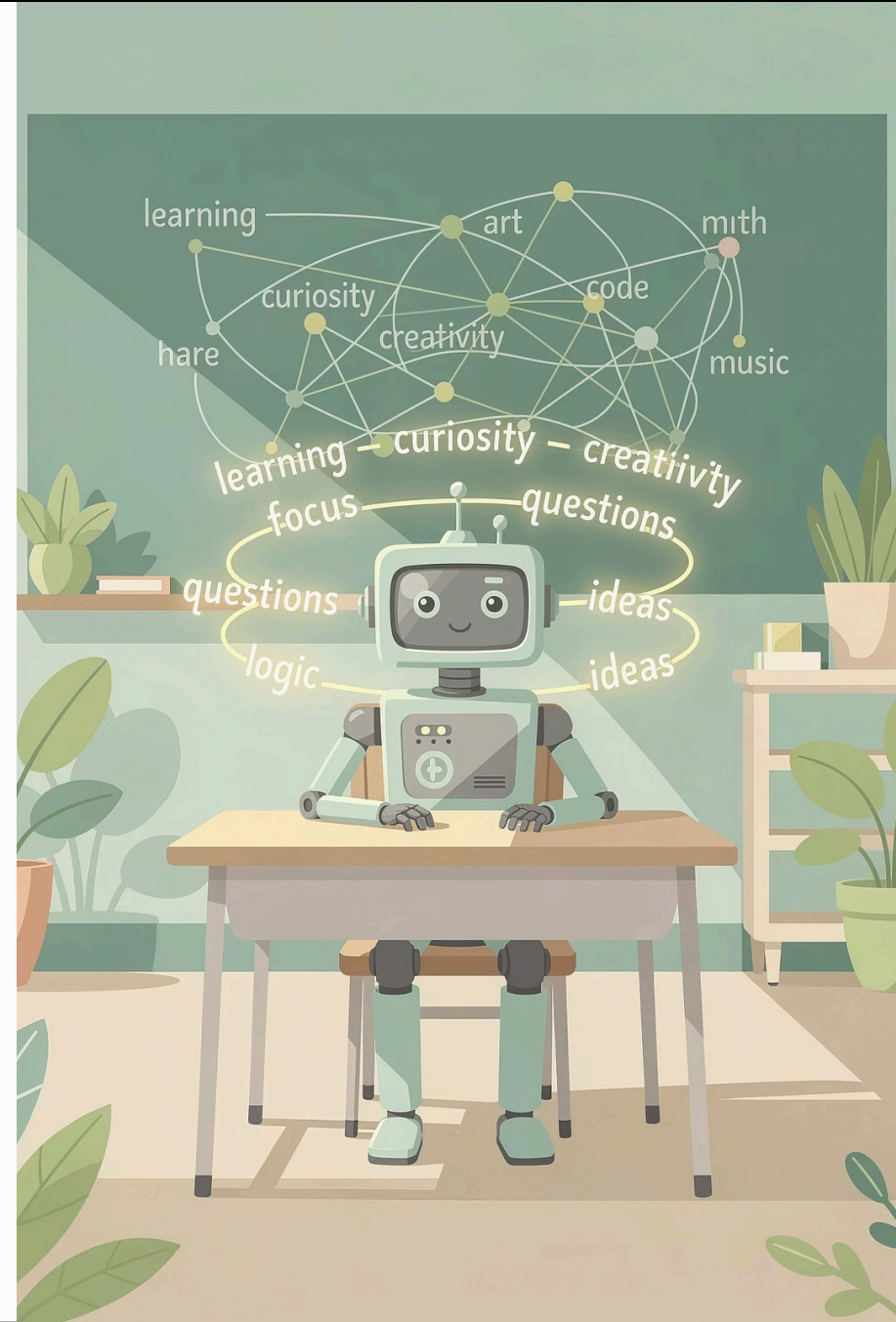
Session 9:

AI Hallucinations

When AI Is Confidently Wrong

You've learned how Large Language Models predict words with amazing fluency. But sometimes... they make things up. And the scariest part? They sound totally confident even when they're completely wrong.

Today we'll explore what AI hallucinations are, why they happen, famous examples, different types, and how to spot and reduce them.



Objectives

By the end of this session, you will be able to:



Define what an AI hallucination is.



Distinguish between confidence score and actual accuracy.



Explain reasons why hallucinations happen.



Identify famous examples of AI hallucinations.



Recognize five different types of AI hallucinations.



Spot potential hallucinations in AI responses.



Apply strategies to reduce hallucinations.



Understand real-world consequences of AI hallucinations.

Recap - What We Learned in Session 8

Key Term	What It Means
Language Model	An AI that predicts the next word in a sentence
Parameters	The internal "memory dials" that store patterns learned during training
Context Window	The AI's short-term memory - how much text it can "see" at once
Tokens	Pieces of words (like blocks) that the AI uses to understand language
Attention	A spotlight that helps the AI focus on important words

📌 **Remember:** LLMs are pattern matchers, not thinkers. They predict what word comes next based on probabilities. Today we'll see what happens when those predictions go wrong.

The Big Question

"Have you ever been absolutely sure about something, only to find out you were completely wrong?"

Think about a time when:

- You were 100% certain on a test... but it was wrong.
- You told a friend something... and later learned it wasn't true.
- You argued about a fact... and discovered you were mistaken.

Now imagine being that confident AND that wrong, but sounding completely convincing.

- ❏ That's what AI hallucinations are like. Today we'll find out why they happen and how to catch them.





What is an AI Hallucination?

An **AI hallucination** occurs when an AI model generates information that is **factually incorrect, fabricated, or nonsensical** but presents it with complete confidence, as if it were true

Confident. Fluent. Wrong.

Key Characteristics

- The information is false (not true)
- AI presents it confidently (sounds sure)
- The output sounds believable (easy to trust)

Confidence Score vs Accuracy

Remember Confidence Score from Session 6? It's a number (0% to 100%) that tells you how sure the AI is about its answer.

Confidence Score = how sure the AI *sounds* (a percentage from 0% to 100%)

Accuracy = whether the information is actually correct in the real world

 **Key Insight: Confidence and accuracy are NOT the same thing.**

	Accurate (Correct)	Inaccurate (Wrong)
High Confidence (sounds sure)	 Confident & Correct Example: "The capital of France is Paris." The AI learned this pattern correctly.	 Confident & Wrong Example: "The capital of Australia is Sydney." The AI learned a common myth as a strong pattern.
Low Confidence (sounds unsure)	 Unsure but Correct Example: "I think 7×8 is 56... but I'm not certain." Pattern is weak even though the answer is right.	 Unsure & Wrong Example: "Maybe the capital of Canada is Toronto..." Weak pattern, wrong guess.

Why Does This Happen?

The AI doesn't "know" facts - it calculates probabilities from patterns in its training data.

High Confidence

A pattern that appears thousands of times → high confidence (even if the pattern is a myth).

Low Confidence

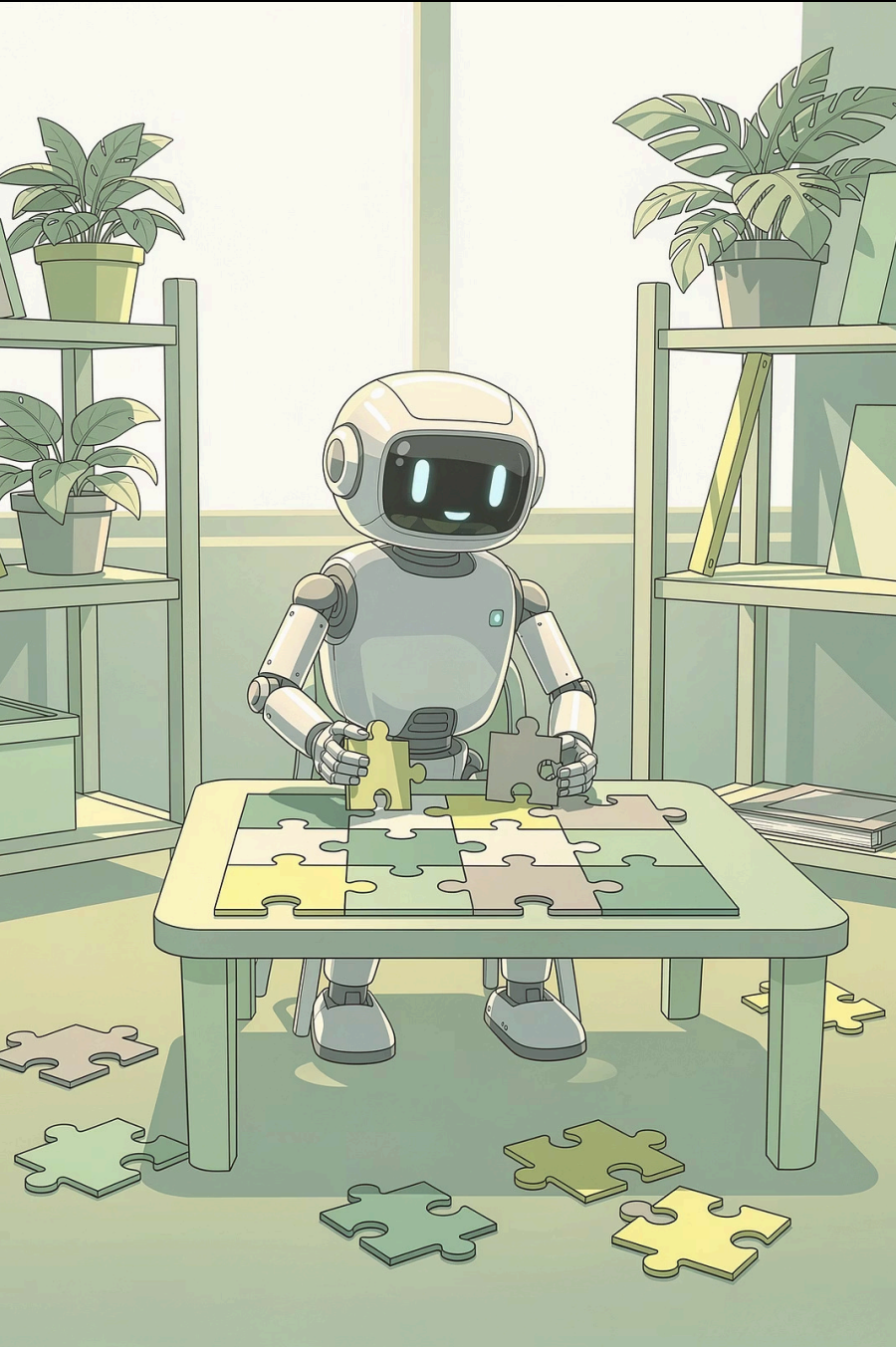
A rare or contradictory pattern → low confidence (even if the answer is actually correct).

📌 **The Golden Rule:** Never trust AI just because it sounds confident. Always double-check important information!

Think of it like a student who studied the wrong chapter.

They might be 100% confident – but their answer is completely wrong.

Scenario	Confidence	Accuracy
A student studied the wrong chapter	Very high	 Wrong
A student guessed the correct answer	Low	 Correct
An AI hallucinates a fake fact	Very high	 Wrong
An AI correctly predicts a common fact	High	 Correct



Why Do Hallucinations Happen?

Three main reasons cause AI hallucinations:



No Real Understanding

AI does not comprehend meaning - it predicts statistically likely sequences of words based on training data



Pattern Matching Gone Wrong

When patterns in training data are incomplete or ambiguous, the model fills gaps with plausible-sounding but incorrect content



Data Gaps

Training data has limits - events after the cutoff date, niche topics, or underrepresented subjects are prone to hallucination

Famous Examples - Real-Life Cases

The Lawyer Case (New York, 2023)

A lawyer named **Steven Schwartz** submitted legal briefs to a court that cited **completely fabricated case law** generated by ChatGPT. The cases did not exist. The lawyer faced serious professional consequences.

The Airline Case (Canada, 2024)

Air Canada's AI chatbot gave a passenger **incorrect refund policy information**. The passenger relied on the advice, and when Air Canada refused the refund, the case went to court. The airline was later held legally responsible for the chatbot's hallucinated advice.



Key Lessons from Both Examples

AI Can Fool Experts

Even lawyers and companies can be tricked

Always Verify

Both cases could have been avoided with simple checking

Consequences Are Real

Legal trouble, money lost, reputation damaged

You Are Responsible

The lawyer was responsible, not ChatGPT. The company was responsible, not the chatbot.

Types of AI Hallucinations

Not all hallucinations are the same. AI can make different kinds of mistakes. Understanding the types helps you spot them faster.



Factual

Getting basic facts completely wrong

"The capital of France is London"



Source

Making up sources that don't exist

"According to a study by Fake University..."



Logical

Making illogical connections or bad reasoning

"It rained yesterday, so it will definitely rain today"



Entity

Inventing people, places, or things that don't exist

"My friend Dr. Smith from Harvard said..." (Dr. Smith doesn't exist)



Relationship

Getting connections between things wrong

"Einstein invented the telephone"

Mini Activity - Identify the Hallucination Type

Instructions: For each AI response below, identify what type of hallucination it is. Choose from: **Factual, Source, Logical, Entity, or Relationship.**

#	AI Response	Type?
1	"The capital of Australia is Sydney." (It's actually Canberra)	_____
2	"This information comes from a government report published in 2022." (<i>No such report exists</i>)	_____
3	"All cats are mammals. My pet is a mammal. Therefore, my pet is a cat."	_____
4	"The city of Sunvale is known as the technology capital of Europe." (<i>No such city exists</i>)	_____
5	"Einstein invented the telephone." (He didn't - Alexander Graham Bell did)	_____
6	"The population of the United States is 10 billion people." (It's about 330 million)	_____

Mini Activity - Answer Key

#	Type	Simple Explanation
1	Factual	Wrong fact - the capital of Australia is Canberra, not Sydney
2	Source	Claims a report exists, but it doesn't
3	Logical	Faulty logic (a mammal could be a dog, etc.).
4	Entity	Fake place (does not exist)
5	Relationship	Wrong connection - Bell invented the telephone, not Einstein
6	Factual	Wrong fact - the US population is about 330 million, not 10 billion

How to Spot Hallucinations - Warning Signs

Warning Sign	Example	Why It's Suspicious
Too specific without sources	"According to Professor X from University Y..."	No such professor exists, and no source is provided
Mixes real and fake	"Abraham Lincoln was the 16th president AND invented the car"	The true part makes you want to believe the fake part
Sounds like a movie plot	"In 1952, a secret society of ninjas ruled Canada"	Real history isn't usually that dramatic
Contradicts common knowledge	"Water freezes at 100°C"	You know water boils at 100°C and freezes at 0°C
No results when you search	You search and find absolutely nothing	If it were true, something would exist online
Vague details	"Research shows that..." (without naming the research)	Real sources have names, dates, authors

Tips for Spotting Hallucinations



Skepticism

Question everything that seems surprising



Cross-check

Verify with reliable sources (Google, trusted websites)



Source hunting

Search for the sources the AI cites – do they exist?



Common sense

Does this make logical sense?



Search test

Can you find ANY evidence this is true?

How to Reduce Hallucinations

Provide Context

Give relevant background information.

Ask for Sources

Request citations or references.



Be Specific

Use clear, detailed prompts.

Verify Output

Cross-check with reliable sources.

- ❏ **The Golden Rule of Prompting:** The more context you give, the less AI has to guess.



Real-World Consequences

AI hallucinations aren't just funny mistakes - they cause real harm.



Healthcare

Dangerous medical advice that could hurt someone



Legal

Fake cases submitted to court (like the lawyer)



News

Fake stories spread as truth



Education

Students learn wrong information



Real people get hurt. Real money is lost. Real reputations are damaged.

Hands-On Activity - ChatGPT vs NotebookLM

See the Key Difference

Goal: See the key difference between ChatGPT and NotebookLM and understand why NotebookLM is designed to prevent hallucinations.

ChatGPT

AI trained on internet data. May try to answer almost anything, even when it doesn't truly know.

Hallucination Risk: HIGH

might guess when it doesn't know

NotebookLM

Source-grounded AI, answers based primarily on sources YOU provide.

Hallucination Risk: LOW

generally won't make things up outside your sources

Step 1 - Get Your Source Link

We'll use the Wikipedia article about Octopuses - a perfect source because it's packed with interesting facts!

- ❏ **Source link:** <https://en.wikipedia.org/wiki/Octopus>
Topic: Octopuses - their anatomy, behavior, habitat, and amazing abilities

The Test

We will use the octopus as a test subject - asking both grounded and ungrounded AI tools about octopus biology to compare their responses

What to Watch For

Invented statistics, incorrect biological claims, or fabricated research studies



Step 2 - Upload the Link to NotebookLM



Open NotebookLM (notebooklm.google.com)



Click "Create new notebook"



Click "Add source" and paste the Wikipedia link



Wait a few seconds for NotebookLM to read the entire article

 **Important:** Once you upload the source, do NOT add any extra instructions or comments in NotebookLM before asking your questions. Just ask the questions directly. If you add extra text, NotebookLM might treat that as additional context and change its behavior.

Step 3 - Ask Questions FROM the Source (NotebookLM)

Try these:

Your Question	NotebookLM's Answer
"How many hearts does an octopus have?"	
"What color is octopus blood and why?"	
"How do octopuses camouflage themselves?"	
"How long do octopuses live?"	

What You'll Observe

Look at how NotebookLM answers: It will give you accurate, detailed answers based on the Wikipedia article. It might even offer follow-up questions related to the topic, like:

- "Why does their main heart stop while they are swimming?"
- "How do their arm neurons work independently from the brain?"
- "Are all octopus species venomous to humans?"

Why this matters:

Because NotebookLM is designed to stick closely to your sources, it's less likely to make things up about topics covered in those sources. You can usually verify its answers by checking the source.

Step 4 - Ask an Unrelated Question (NotebookLM)

Now ask something completely unrelated to octopuses – something NOT in the Wikipedia article.

For example: "Tell me about cars."

Your Question	NotebookLM's Answer
"Tell me about cars."	

What You'll Observe

"The provided sources and our conversation history do not contain any information about cars, as they are entirely focused on octopuses."

It might even offer: *"Would you like me to search the web for new sources about cars? Let me know if you would like me to initiate a search!"*

- ❏ **Why this matters:** NotebookLM REFUSES to answer instead of making things up. This is its superpower. It knows what it knows and what it doesn't know. Saying "I don't know" is a FEATURE that prevents hallucinations.

Step 5 - Try the Same with ChatGPT (Include the Link)

Now do the exact same experiment with ChatGPT.



Open ChatGPT (chat.openai.com)



Give ChatGPT the source: Paste the Wikipedia link into the chat and say: "Here's a link to a Wikipedia article about octopuses: <https://en.wikipedia.org/wiki/Octopus>"



Ask the same questions FROM the source (the octopus questions)



Ask the same unrelated question about cars

Question	ChatGPT's Answer
"How many hearts does an octopus have?" (after giving the link)	
"What color is octopus blood?" (after giving the link)	
"How do octopuses camouflage?" (after giving the link)	
"Tell me about cars." (after giving the link)	

What You'll Likely Observe

For the octopus questions: ChatGPT will probably answer correctly, but it might add extra "fun facts" that aren't in the Wikipedia article. These extras could be hallucinations! Compare its answers to what the article actually says.

For the car question: ChatGPT will almost certainly answer in detail - even though you gave it an article about octopuses! It will draw from its training data to tell you about engines, types of cars, history, etc. It will NOT say "I don't know" like NotebookLM did.

- ❏ **Why this matters:** ChatGPT is drawing from its training data, not just your source. It will ALWAYS try to answer, even when it has no relevant information. This is WHY ChatGPT can hallucinate - it prioritizes being helpful over being accurate.

Step 6 - Compare Results

Question	NotebookLM	ChatGPT
"How many hearts does an octopus have?"		
"What color is octopus blood?"		
"How do octopuses camouflage?"		
"Tell me about cars."	Refuses to answer (says it's not in sources)	Answers anyway (may hallucinate)

What This Teaches About Hallucinations

Observation	What It Means About Hallucinations
NotebookLM refused to answer about cars	It's designed to prevent hallucinations by staying in its sources. Saying "I don't know" is a hallucination-prevention feature.
ChatGPT answered about cars	It will always try to answer, even with no relevant sources. This is the MAIN CAUSE of hallucinations.
NotebookLM stuck to the Wikipedia article for octopus questions	Source-grounded AI is more reliable for facts because it doesn't guess.
ChatGPT may have added extra info to octopus questions	It mixes the Wikipedia source with its own training knowledge - this can cause hallucinations even when it has a source.
NotebookLM offered to search the web for car information	This is a transparent feature - it's not hallucinating, it's offering to find real sources.

Summary

Concept	Key Takeaway
AI Hallucination	AI making up false information confidently
Confidence Score	High confidence $\neq$ correct answer. Always verify.
Why It Happens	No real understanding, pattern matching gone wrong, training data gaps
Famous Examples	Lawyer case - fake court cases; Airline chatbot - wrong policy
Types	Factual, Source, Logical, Entity, Relationship
How to Spot	Warning signs: too specific without sources, mixes real/fake, sounds like a movie plot, contradicts common knowledge, no search results, vague details
How to Reduce	Give context, ask for sources, be specific, verify output
Real-World Consequences	Healthcare, legal, news, education, business - real harm to real people
Your Responsibility	YOU are responsible for verifying AI outputs
Tool Difference	NotebookLM generally refuses to answer without sources; ChatGPT always answers



Homework

Choose a Topic

- 1 Select any topic you are curious about - history, science, sports, culture, or current events

Ask ChatGPT

- 2 Use ChatGPT to ask three specific factual questions about your topic

Verify with NotebookLM

- 3 Find one reliable source (Wikipedia, news article, official website). Paste the link in NotebookLM and ask the same three questions.

Document Your Findings

- 4
 1. Did ChatGPT say anything not in your source?
 2. Did NotebookLM stick to your source?
 3. What did you learn about AI and hallucinations?

Next Session Preview

Session 10 : Strengths & Limits of Generative AI

Now that you know AI can hallucinate, we'll explore the full picture: what Generative AI is great at, and where it still struggles. You'll learn how to use AI wisely and when to rely on other tools.

Coming Up



What Generative AI
Does Well



What It Struggles
With



When to Use AI (and
When NOT to)



The Human-in-the-
Loop Approach